Unit-3

Prompt Engineering

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Syllabus

- **Prompt Engineering: Introduction to Prompt Engineering , The Basic Ingredients of a Prompt , Instruction-Based Prompting , Advanced Prompt Engineering , The Potential Complexity of a Prompt , In-Context Learning: Providing Examples , Chain Prompting: Breaking up the Problem , Reasoning with Generative Models , Chain-of-Thought: Think Before Answering , Self-Consistency: Sampling Outputs, Tree-of-Thought: Exploring Intermediate Steps , Output Verification , Providing Examples , Grammar: Constrained Sampling. Prompt design and Optimization.**

Comparison of Top LLMs

https://lmarena.ai/?leaderboard



- At their core, LLMs are capable of one thing—completing text. The input into the model is called the
- **Prompt**—it is a document, or block of text, that we expect the model to complete. Prompt engineering, then, in its simplest form, is the practice of crafting the prompt so that its completion contains the information required to address the problem at hand.

- Prompt engineering is the process of discovering prompts that reliably yield useful or desired results.
- A prompt is the input you provide, typically text, when interfacing with an AI model like ChatGPT or Midjourney.
- The prompt serves as a set of instructions the model uses to predict the desired response.
- Text from large language models (LLMs) like ChatGPT, or images from diffusion models like Midjourney.

The Five Principles of Prompting

1.Give Direction

Describe the desired style in detail, or reference a relevant persona

2.Specify Format

Define what rules to follow, and the required structure of the response

3.Provide Examples

Insert a diverse set of test cases where the task was done correctly

4.Evaluate Quality

Identify errors and rate responses, testing what drives performance.

5.Divide Labor

Split tasks into multiple steps, chained together for complex goals

Prompt Engineering

- Prompt engineering is more than designing effective prompts.
- It can be used as a tool to evaluate the output of a model as well as to design safeguards and safety mitigation methods.
- This is an iterative process of prompt optimization and requires experimentation.
- There is not and unlikely will ever be a perfect prompt design.

The Basic Ingredients of a Prompt

- **An LLM is a prediction machine.**
- **Based on a certain input, the prompt, it tries to predict the words that might follow it.**
- **At its core , the prompt does not need to be more than a few words to elicit a response from the LLM.**

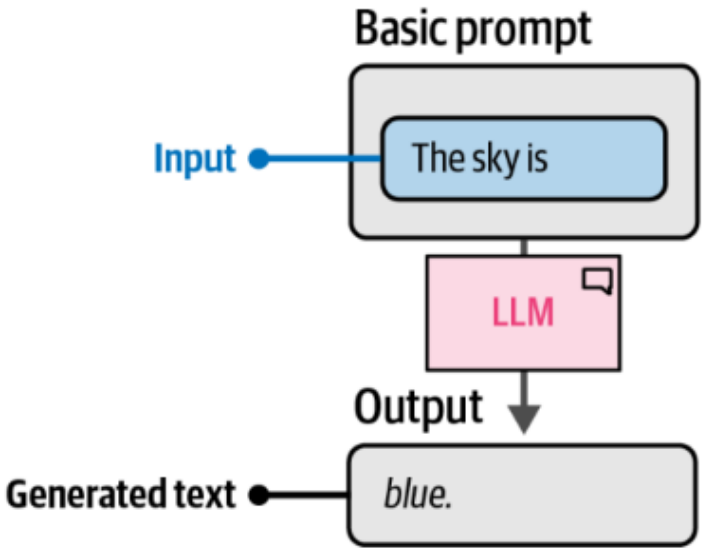


Figure 6-6. A basic example of a prompt. No instruction is given so the LLM will simply try to complete the sentence.

Two components of a basic instruction prompt

- **we generally approach prompt engineering by asking a specific question or task the LLM should complete.**
- **To e[illegible]pt.**

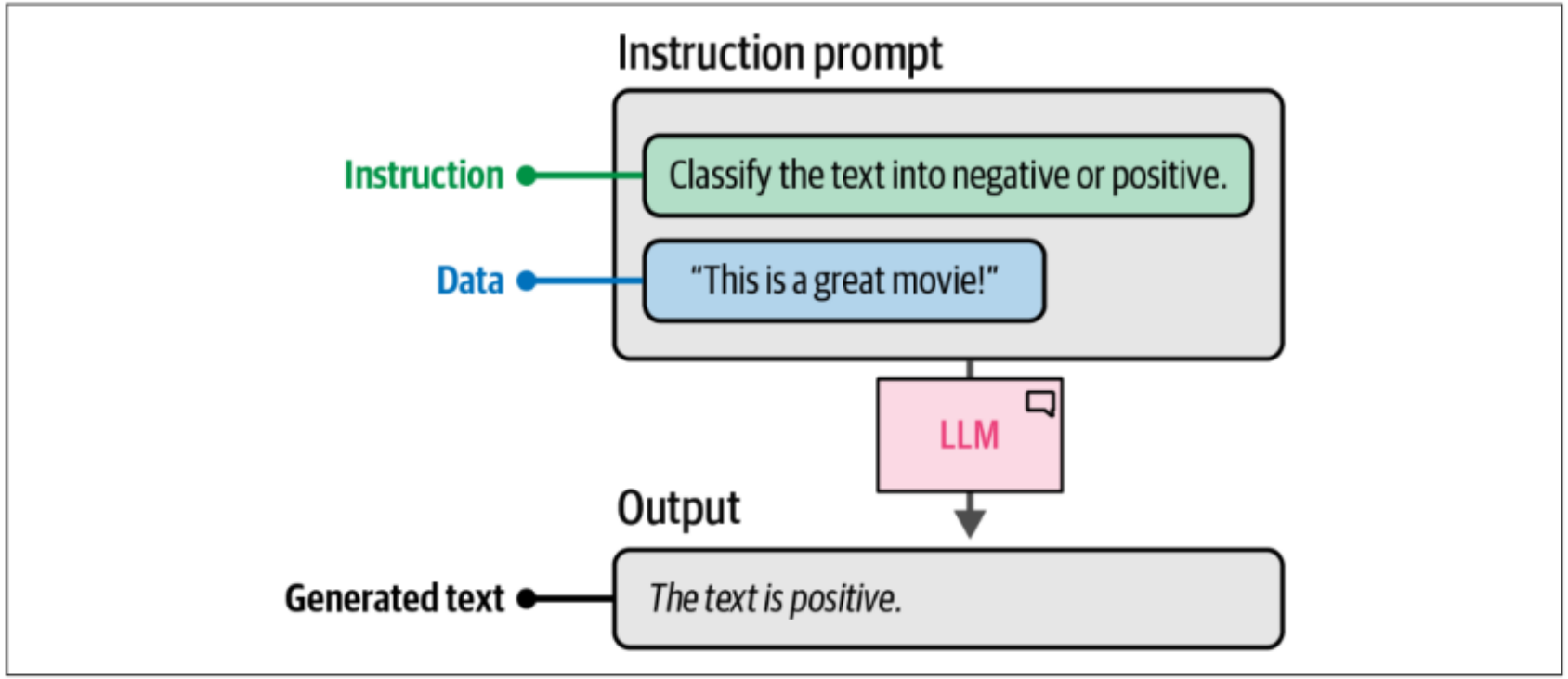


Figure 6-7. Two components of a basic instruction prompt: the instruction itself and the data it refers to.

- More complex use cases might require more components in a prompt.
- For instance, to make sure the model only outputs “negative” or “positive” we can introduce output indicators that help guide the model.
- We prefix the sentence with “Text:” and add “Sentiment:”
- To prevent the model from generating a complete sentence. Instead, this structure indicates that we expect either “negative” or “positive.”
- Although the model might not have been trained on these components directly, it was fed enough instructions to be able to generalize to this structure.

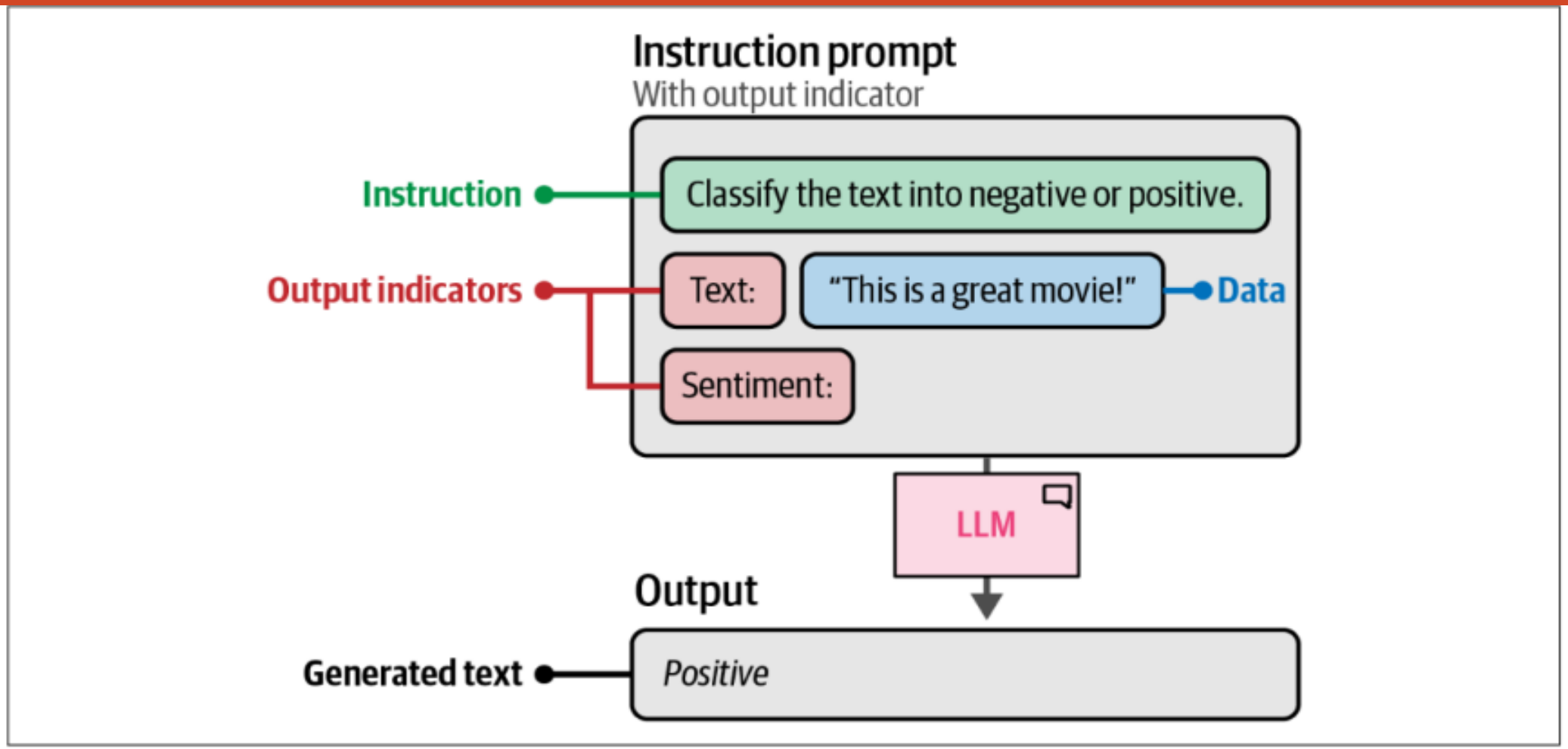


Figure 6-8. Extending the prompt with an output indicator that allows for a specific output.

Instruction-Based Prompting

- **Although prompting comes in many flavors, from discussing philosophy with the LLM to role-playing with your favorite superhero, prompting is often used to have the LLM answer a specific question or resolve a certain task.**
- **This is referred to as instruction-based prompting.**

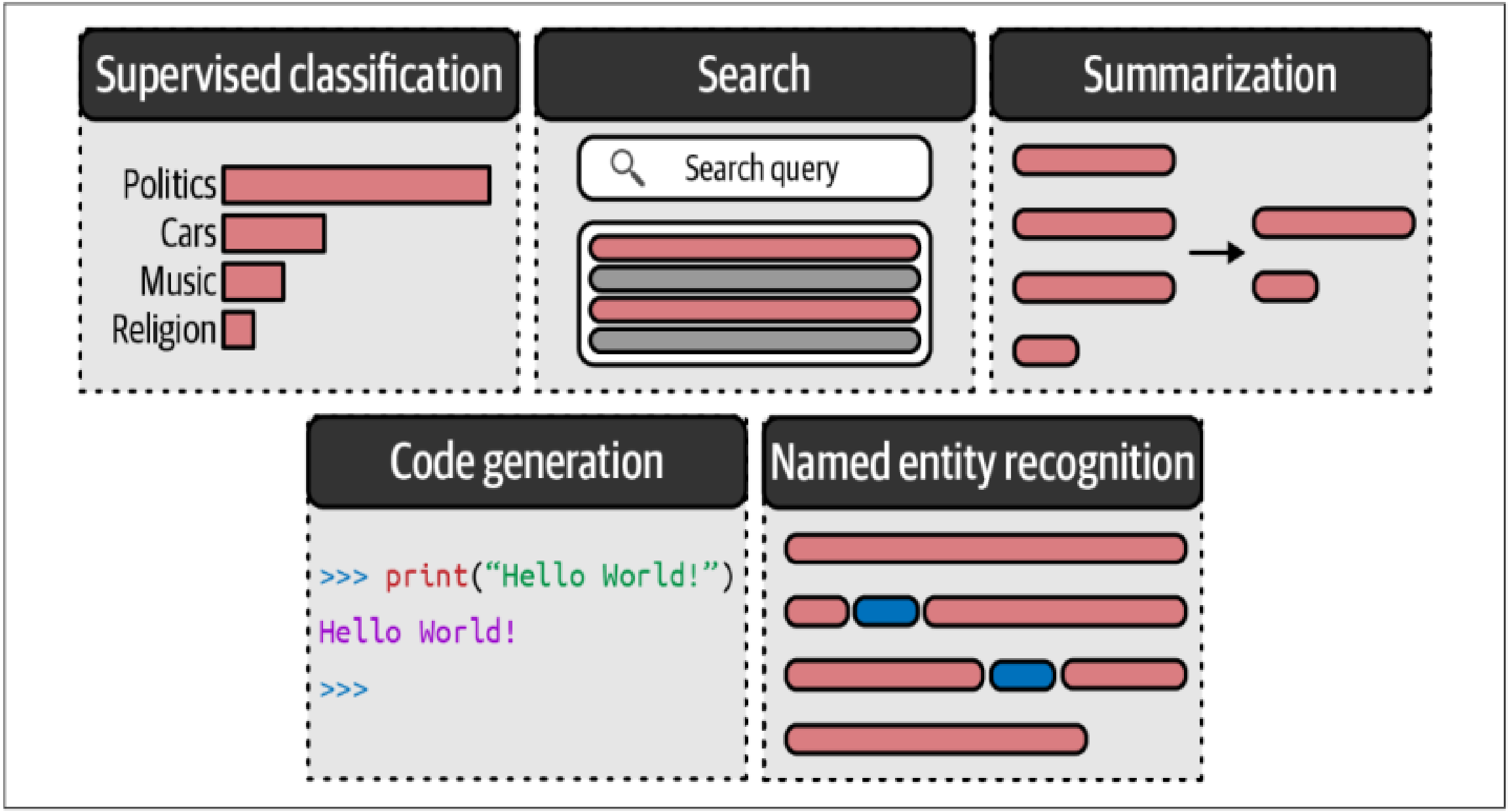


Figure 6-9. Use cases for instruction-based prompting.

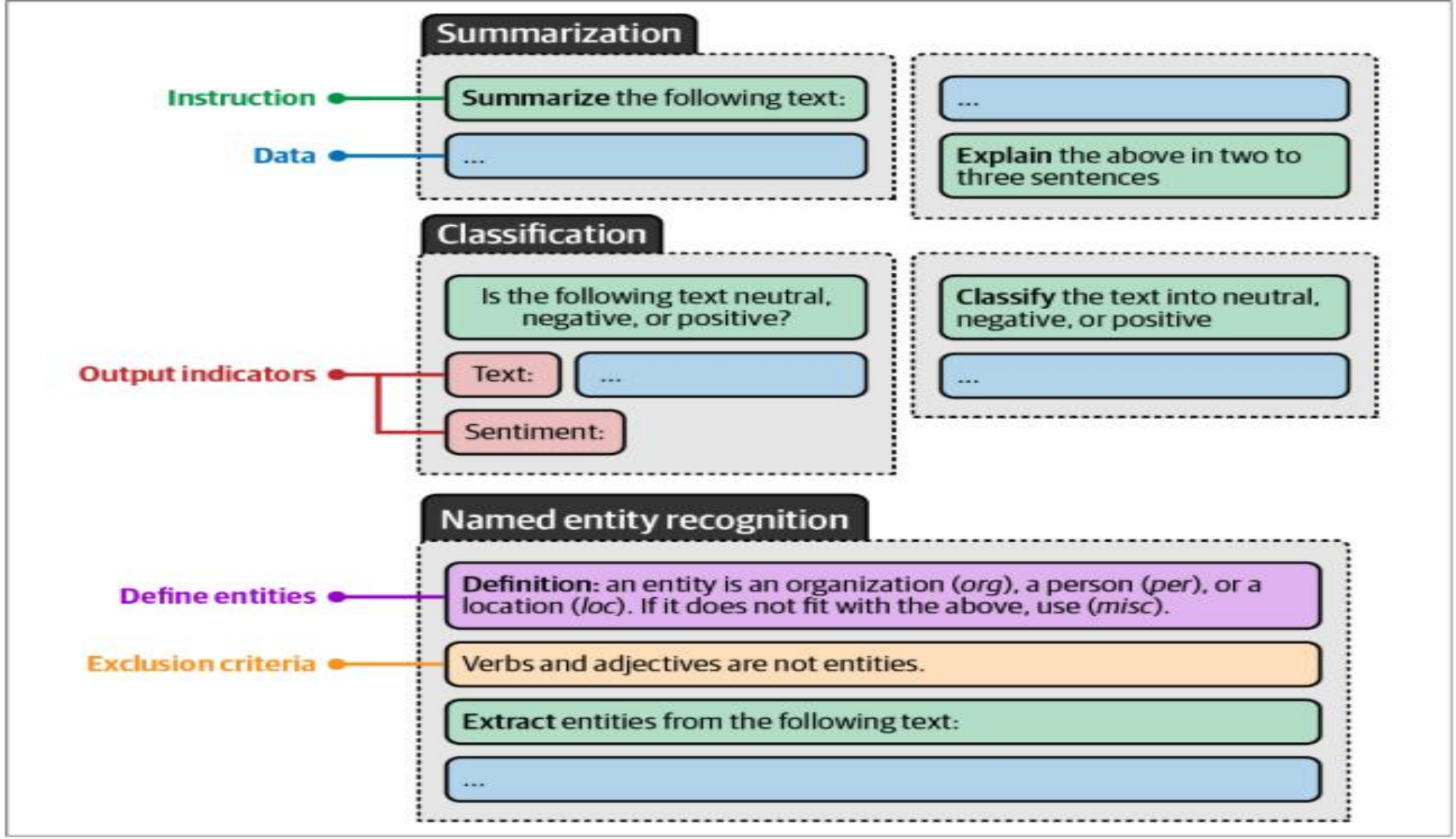


Figure 6-10. Prompt examples of common use cases. Notice how within a use case, the structure and location of the instruction can be changed.

Specificity

- Accurately describe what you want to achieve. Instead of asking the LLM to “Write a description for a product” ask it to “Write a description for a product in less than two sentences and use a formal tone.”

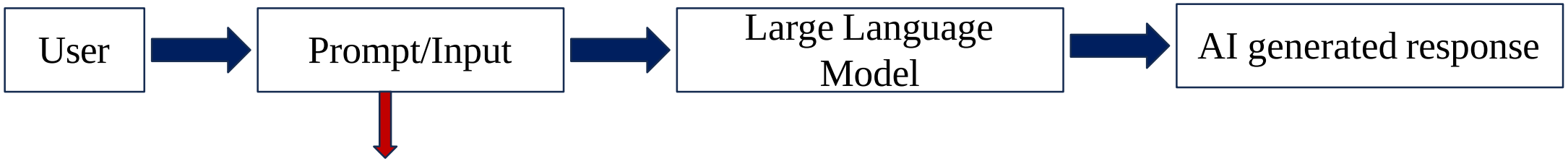
Hallucination

- LLMs may generate incorrect information confidently, which is referred to as hallucination. To reduce its impact, we can ask the LLM to only generate an answer if it knows the answer. If it does not know the answer, it can respond with “I don’t know.”

Order

- Either begin or end your prompt with the instruction. Especially with long prompts, information in the middle is often forgotten.
- LLMs tend to focus on information either at the beginning of a prompt (primacy effect) or the end of a prompt (recency effect).

Concept of Prompt Engineering



- ✓ Prompt engineering is the process of designing effective inputs for LLMs.
- ✓ Better prompts lead to better responses.

A prompt may include

- Instructions
- Questions
- Context
- Constraints
- Examples

Example
Summarize this article in five bullet points.

Why Prompts Matter?

Define the task	The prompt tells the model whether to summarize, translate, explain, classify, generate code, or answer a question.
Provide context	Additional background information helps the model understand the situation and produce more relevant responses.
Specify the audience	Indicating the intended audience allows the model to adjust the complexity and terminology.
Control the output format	The prompt can request bullet points, tables, JSON, code, or essays, making the response easier to use.
Reduce ambiguity	Clear and specific instructions minimize misunderstandings and improve the accuracy of the response

Prompts

Characteristics of Good Prompts

- Clear
- Context-rich
- Unambiguous
- Goal-oriented
- Specific

Things to avoid:

- Vague instructions
- Missing context
- Multiple unrelated questions

Vague Prompt

Prompt:

Explain machine learning.

Possible Response

Machine learning is a branch of artificial intelligence that enables computers to learn from data without being explicitly programmed. It includes supervised learning, unsupervised learning, reinforcement learning, and many algorithms such as decision trees, neural networks, and support vector machines.

Issue:

The response is broad and may be too detailed or too technical for the intended audience.

Specific Prompt

Prompt:

Explain machine learning to a first-year engineering student in 150 words. Use simple language, include one real-world example

Possible Response

Machine learning is a type of artificial intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every task. For example, an email application can learn to identify spam messages by analyzing thousands of previous emails. As it processes more data, it becomes better at distinguishing spam from legitimate emails.

Key Points
Learns patterns from data
Improves performance with experience
Used in applications such as spam filtering, recommendation systems, and image recognition

Why it's better:

The response matches the audience, length, and format requested.

Techniques of Prompting

- ✓ Zero shot Prompting
- ✓ One shot Prompting
- ✓ Few shot prompting
- ✓ Chain of thoughts.
- ✓ Structured Output Prompting
- ✓ Role Prompting

Zero Shot Prompting

- ✓ A prompting technique where a Large Language Model (LLM) performs a task without being provided with any examples.
- ✓ The model relies solely on the instructions in the prompt and its pre-trained knowledge to generate the response.

How it works

- ✓ No training examples are included in the prompt. The prompt contains only the task description or question.
- ✓ The model infers the required output based on its learned knowledge.

Example

Prompt:
Classify the sentiment of the following sentence as Positive, Negative, or Neutral:
"The movie had stunning visuals, but the storyline was disappointing."

LLM Output:
Negative
(The model correctly identifies the overall sentiment without seeing any prior examples.)

One Shot Prompting

- One-shot prompting is a prompting technique where a Large Language Model (LLM) is provided with exactly one example of the task before being asked to perform it on a new input.

- The single example helps the model understand the expected format and behavior.

How it works

- ✓ Provide one input–output example.
- ✓ The model identifies the pattern from the example.
- ✓ It applies the same pattern to the new input.

Example

Prompt:
Sentence: "The customer support was excellent."
Sentiment: Positive

Sentence: "The laptop battery drains very quickly."
Sentiment:

LLM Output:
Negative
(The model learns the sentiment classification task from a single example and applies it to the new sentence.)

Few Shot Prompting

- ✓ Few-shot prompting is a prompting technique where a Large Language Model (LLM) is given a small number of input–output examples before the actual task.
- ✓ These examples help the model understand the desired format, style, and reasoning pattern, leading to more accurate responses.

How it works?

- ✓ Provide 2–5 examples demonstrating the task.
- ✓ The model learns the pattern from the examples.
- ✓ It then applies the same pattern to the new input.

Example

Prompt:
Sentence: "I absolutely loved the food."
Sentiment: Positive

Sentence: "The service was very slow."
Sentiment: Negative

Sentence: "The phone has a good camera but poor battery life."
Sentiment:

Applications of zero-shot, one-shot, and few-shot prompting

Zero Shot prompting

- ✓ Text summarization
- ✓ Translation
- ✓ Sentiment analysis
- ✓ Question answering
- ✓ Grammar correction

One shot prompting

- ✓ Teaching a specific response format
- ✓ Maintaining a writing style
- ✓ Explaining custom labeling
- ✓ Generating structured outputs

Few shot prompting

- ✓ Custom text classification
- ✓ Information extraction
- ✓ Domain-specific tasks
- ✓ Reasoning tasks
- ✓ Intent classification

Chain-of- Thought (CoT) Prompting

- ✓ Chain-of-Thought (CoT) prompting is a technique in which the model is encouraged to reason through a problem step by step before producing the final answer.
- ✓ Instead of giving only the answer, the model breaks the problem into intermediate reasoning steps, making it more accurate for complex reasoning, mathematics, logic, and decision-making tasks.

Prompt

A shop sells a notebook for $5 and a pen for $2.
A customer buys 3 notebooks and 4 pens.
Calculate the total cost. Think step by step.



LLM Response

Step 1: Cost of 3 notebooks = 3 × $5 = $15
Step 2: Cost of 4 pens = 4 × $2 = $8
Step 3: Total cost = $15 + $8 = $23

Final Answer: $23

Role Prompting

Role prompting is a technique where the Large Language Model (LLM) is assigned a specific role, profession, or persona before answering a question.

By defining a role (e.g., teacher, doctor, software engineer, financial advisor), the model adapts its tone, level of detail, vocabulary, and perspective to suit that role.

Prompt

You are an experienced physics professor teaching first-year engineering students.

Explain Newton's Second Law in simple language with a real-life example.

LLM Response

Newton's Second Law states that the force acting on an object equals its mass multiplied by its acceleration (F = ma).

For example, pushing an empty shopping cart is easier than pushing a fully loaded cart. The heavier cart has greater mass, so it requires more force to achieve the same acceleration.



Structured Output Prompting

Structured Output Prompting is a prompting technique where the user specifies the exact format in which the Large Language Model (LLM) should return its response.

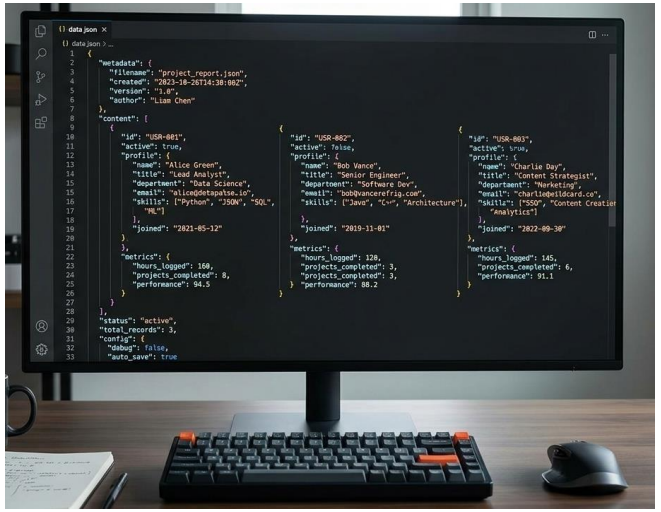
Prompt

Extract the information from the paragraph below and return
the output as a JSON object.

Text:
"John Smith is a Data Scientist at ABC Technologies.

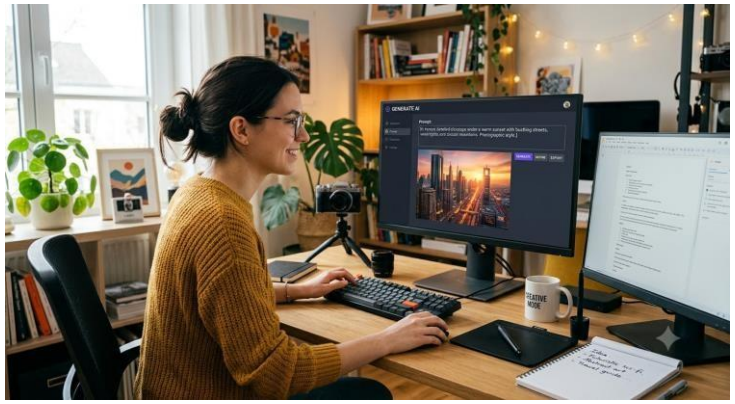
LLM Response

```
{
 "Name": "John Smith",
 "Company": "ABC Technologies",
 "Designation": "Data Scientist",
 "Experience": "5 years",
 "Specialization": "Machine Learning"
}
```

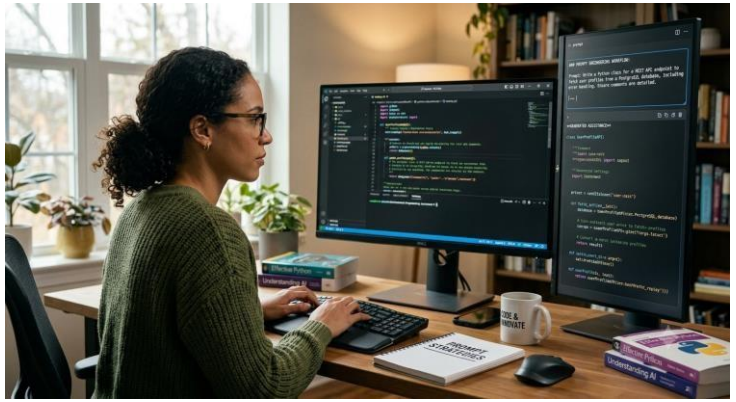


Applications of Prompt Engineering

- ✓ Content generation
- ✓ Data analysis
- ✓ Programming assistance
- ✓ Education
- ✓ Healthcare
- ✓ Customer support
- ✓ Research
- ✓ Document summarization
- ✓ Translation









Advanced Prompt Engineering

Prompting can grow complex quite quickly and as a result is an often-underestimated component of leveraging LLMs.

The Potential Complexity of a Prompt

Persona
Describe what role the LLM should take on. For example, use "You are an expert in astrophysics" if you want to ask a question about astrophysics.

Instruction
The task itself. Make sure this is as specific as possible. We do not want to leave much room for interpretation.

Context
Additional information describing the context of the problem or task. It answers questions like "What is the reason for the instruction?"

Format
The format the LLM should use to output the generated text. Without it, the LLM will come up with a format itself, which is troublesome in automated systems.

Audience
The target of the generated text. This also describes the level of the generated output. For education purposes, it is often helpful to use ELI5 ("Explain it like I'm 5").

Tone
The tone of voice the LLM should use in the generated text. If you are writing a formal email to your boss, you might not want to use an informal tone of voice.

Data
The main data related to the task itself.

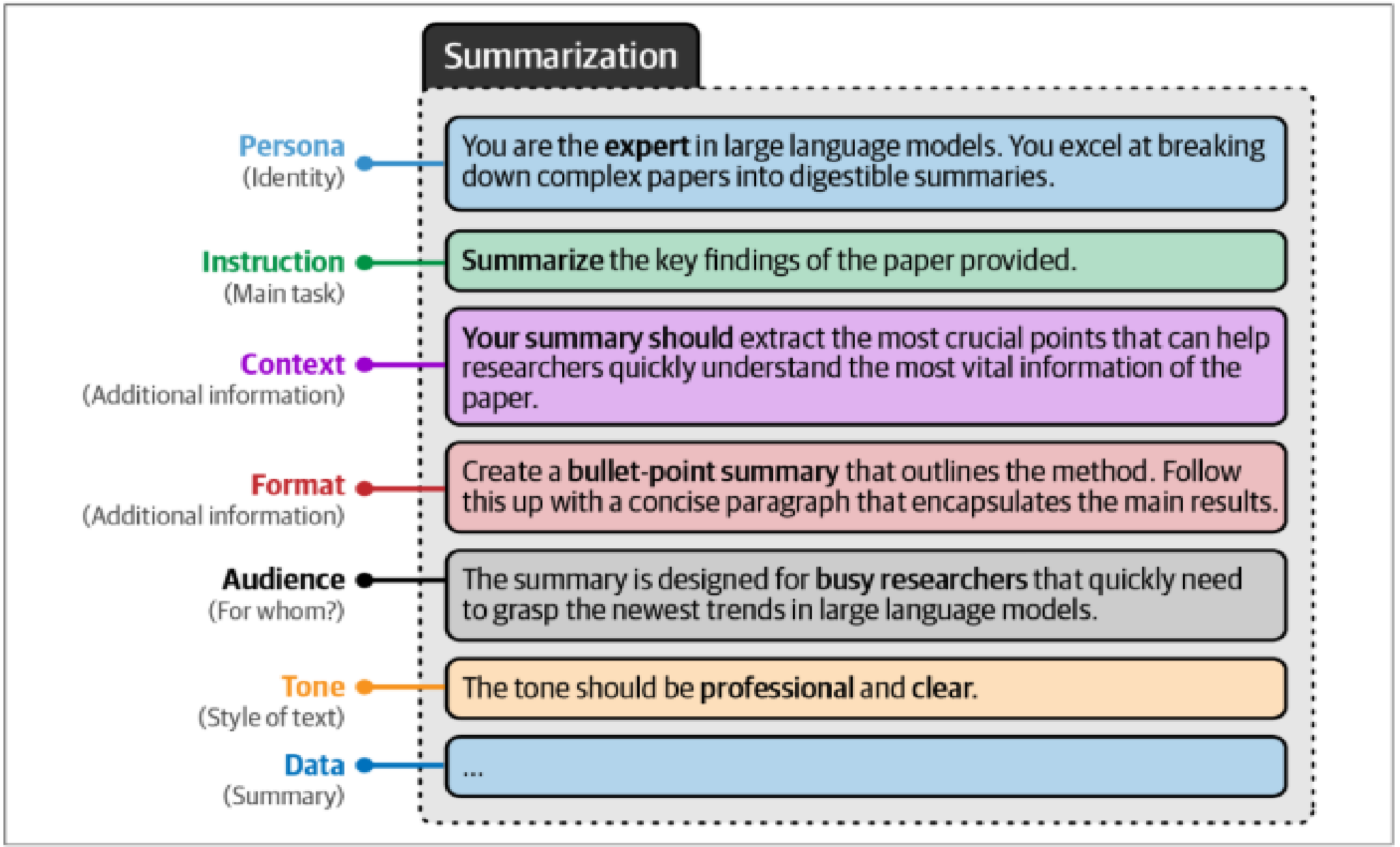


Figure 6-11. An example of a complex prompt with many components.

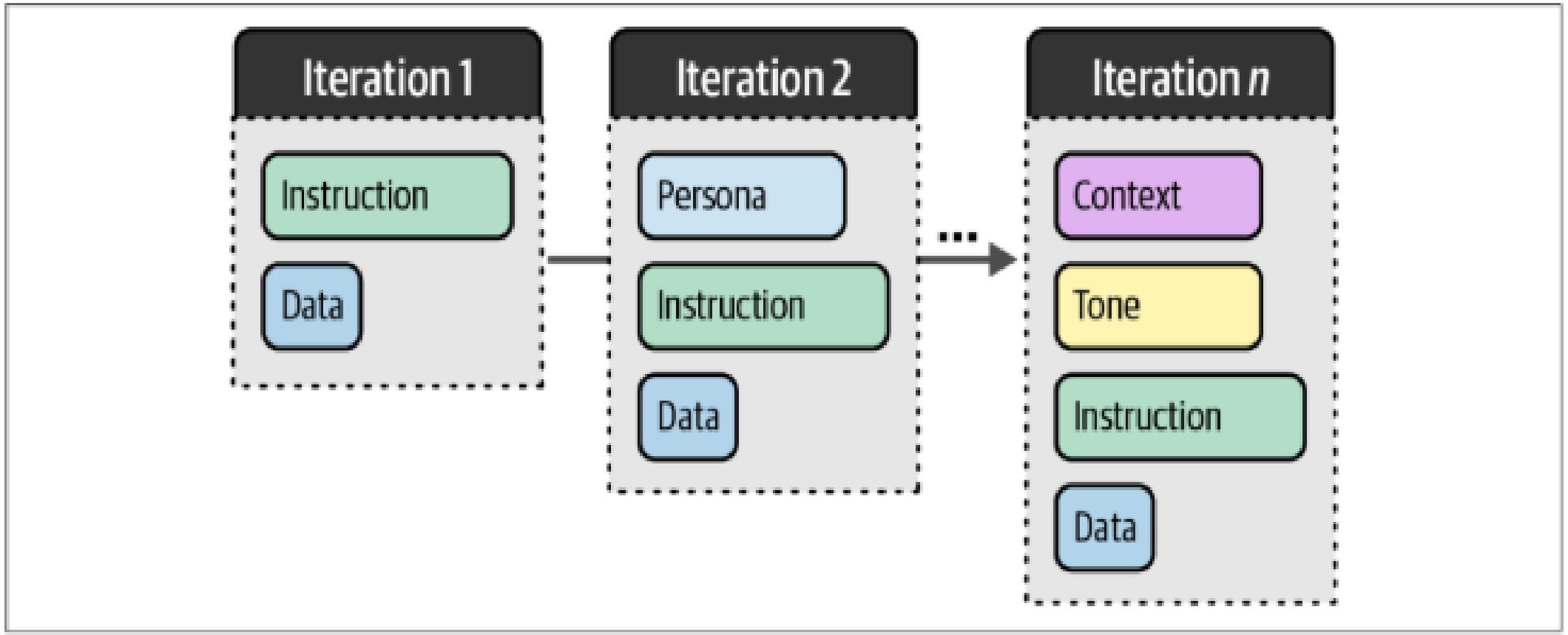


Figure 6-12. Iterating over modular components is a vital part of prompt engineering.

```
# Prompt components
persona = "You are an expert in Large Language models. You excel at breaking
down complex papers into digestible summaries.\n"
instruction = "Summarize the key findings of the paper provided.\n"
context = "Your summary should extract the most crucial points that can help
researchers quickly understand the most vital information of the paper.\n"
data_format = "Create a bullet-point summary that outlines the method. Follow
this up with a concise paragraph that encapsulates the main results.\n"
audience = "The summary is designed for busy researchers that quickly need to
grasp the newest trends in Large Language Models.\n"
tone = "The tone should be professional and clear.\n"
text = "MY TEXT TO SUMMARIZE"
data = f"Text to summarize: {text}"

# The full prompt - remove and add pieces to view its impact on the generated
output
query = persona + instruction + context + data_format + audience + tone + data
```

In-Context Learning: Providing Examples

Zero-shot prompting does not leverage examples, one-shot prompts use a single example, and few-shot prompts use two or more examples.

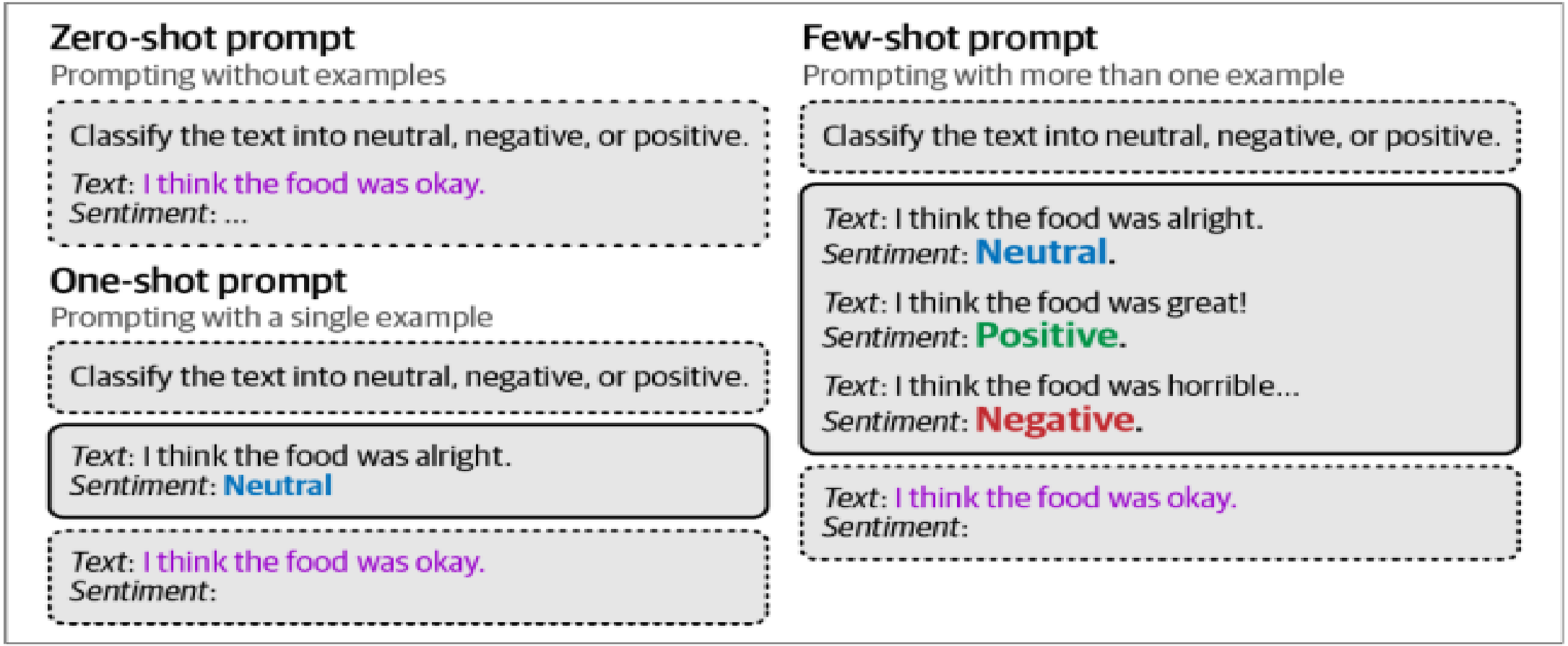


Figure 6-13. An example of a complex prompt with many components.

```
# Use a single example of using the made-up word in a sentence
one_shot_prompt = [
    {
        "role": "user",
        "content": "A 'Gigamuru' is a type of Japanese musical instrument. An
example of a sentence that uses the word Gigamuru is:"
    },
    {
        "role": "assistant",
        "content": "I have a Gigamuru that my uncle gave me as a gift. I love
to play it at home."
    },
    {
        "role": "user",
        "content": "To 'screeg' something is to swing a sword at it. An example
of a sentence that uses the word screeg is:"
    }
]
print(tokenizer.apply_chat_template(one_shot_prompt, tokenize=False))
```

```
<s><|user|>
A 'Gigamuru' is a type of Japanese musical instrument. An example of a sen-
tence that uses the word Gigamuru is:<|end|>
<|assistant|>
I have a Gigamuru that my uncle gave me as a gift. I love to play it at home.<|
end|>
<|user|>
To 'screeg' something is to swing a sword at it. An example of a sentence that
uses the word screeg is:<|end|>
<|assistant|>
```

The prompt illustrates the need to differentiate between the user and the assistant. If we did not, it would seem as if we were talking to ourselves. Using these interactions, we can generate output as follows:

```
# Generate the output
outputs = pipe(one_shot_prompt)
print(outputs[0]["generated_text"])
```

```
During the intense duel, the knight skillfully screeged his opponent's shield,
forcing him to defend himself.
```

Chain Prompting: Breaking up the Problem

- Instead of breaking the problem within a prompt, we can do so between prompts.
- Essentially, we take the output of one prompt and use it as input for the next, thereby creating a continuous chain of interactions that solves our problem.
- We get a sequential pipeline that first creates the product name, uses that with the product features as input to create the slogan, and finally, uses the features, product name, and slogan to create the sales pitch.
- This technique of chaining prompts allows the LLM to spend more time on each individual question instead of tackling the whole problem.

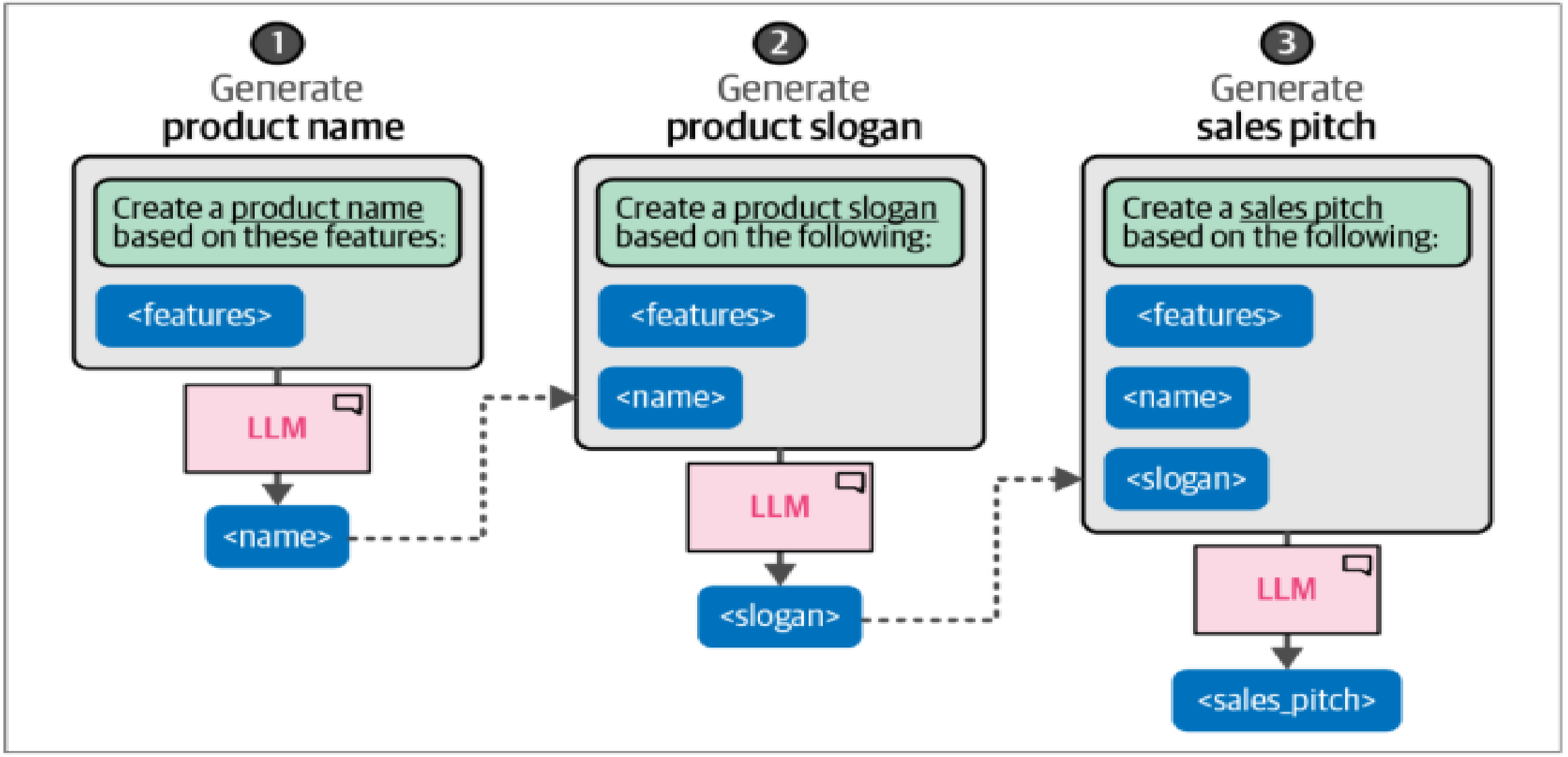


Figure 6-14. Using a description of a product's features, chain prompts to create a suitable name, slogan, and sales pitch.

```
# Create name and slogan for a product
product_prompt = [
    {"role": "user", "content": "Create a name and slogan for a chatbot that
leverages LLMs."}
]
outputs = pipe(product_prompt)
product_description = outputs[0]["generated_text"]
print(product_description)
```

```
Name: 'MindMeld Messenger'

Slogan: 'Unleashing Intelligent Conversations, One Response at a Time'
```

```
# Based on a name and slogan for a product, generate a sales pitch
sales_prompt = [
    {"role": "user", "content": f"Generate a very short sales pitch for the
following product: '{product_description}'"}
]
outputs = pipe(sales_prompt)
sales_pitch = outputs[0]["generated_text"]
print(sales_pitch)
```

```
Introducing MindMeld Messenger - your ultimate communication partner! Unleash
intelligent conversations with our innovative AI-powered messaging platform.
With MindMeld Messenger, every response is thoughtful, personalized, and
timely. Say goodbye to generic replies and hello to meaningful interactions.
Elevate your communication game with MindMeld Messenger - where every message
is a step toward smarter conversations. Try it now and experience the future
of messaging!
```

Response validation

Ask the LLM to double-check previously generated outputs.

Parallel prompts

Create multiple prompts in parallel and do a final pass to merge them. For example, ask multiple LLMs to generate multiple recipes in parallel and use the combined result to create a shopping list.

Writing stories

Leverage the LLM to write books or stories by breaking down the problem into components. For example, by first writing a summary, developing characters, and building the story beats before diving into creating the dialogue.

Prompt is not same as Google Search String

- Input text to these LLMs is known as “Prompt”
- Keep in mind that the quality of the results you receive is directly proportional to the quality of the prompt you provide.
- “Chat-GPT Prompt” is NOT same as “Google Search String”
- Consider the prompt not merely as a query but more as a conversation with Chat GPT.
- Instead of treating it like a Google search, think of it as a robotic assistant that needs clear, specific information to assist you effectively.
- Avoid the brief, small queries or phrases typically used in Google searches
- You need to structure your prompt

Prompt Engineering - Meticulously designing a prompt

- The process of meticulously designing a prompt by incorporating all essential information and relevant elements is termed 'prompt engineering'.
- This involves not only the inclusion of specific instructions and questions but also carefully considering the context, intended tone, and any examples that might guide the model more effectively.
- Effective prompt engineering also requires anticipating potential ambiguities and refining the prompt to be as clear and concise as possible.



You

How to become a data scientist?

ChatGPT

Becoming a data scientist involves a mix of formal education, skill development, and practical experience. Here's a step-by-step guide to help you embark on this career path:

1. **Educational Background**:
   - **Bachelor's Degree**: Ideally in a related field like computer science, statistics, mathematics, or engineering.
   - **Postgraduate Education** (Optional but beneficial): A Master's or Ph.D. in Data Science, Statistics, Machine Learning, or a related field can be very helpful.
2. **Learn Key Skills**:
   - **Programming Languages**: Proficiency in Python or R is essential. SQL is also important for database management.
   - **Statistics and Mathematics**: Understanding probability, statistics, and linear algebra is crucial for data modeling and analysis.
   - **Machine Learning**: Familiarize yourself with basic ML algorithms and concepts.
   - **Data Wrangling**: Learn to clean and preprocess data efficiently.
   - **Data Visualization**: Skills in tools like Tableau, Matplotlib, or Seaborn for communicating data insights.
3. **Practical Experience**:
   - **Projects**: Work on real-world projects or datasets. Kaggle competitions are a good place to start.
   - **Internships**: Gain practical experience and industry knowledge through internships.
4. **Specialize**:
   - Data science is a broad field. You might want to specialize in areas like machine learning, deep learning, natural language processing, or big data technologies.
5. **Soft Skills**:
   - **Communication**: Ability to explain complex results to non-technical stakeholders.
   - **Problem-Solving**: Skills in identifying, analyzing, and solving problems.
   - **Business Acumen**: Understanding the business context for data-driven decision making.
6. **Build a Portfolio**:
   - Showcase your skills and projects on platforms like GitHub. This portfolio is crucial for job applications.
7. **Networking and Community Involvement**:
   - Join data science communities, attend webinars, and participate in hackathons and meetups.
8. **Job Applications**:
   - Start applying for data science positions. Tailor your resume and cover letter for each application, highlighting relevant skills and experiences.
9. **Continuous Learning**:
   - The field of data science is ever-evolving. Stay updated with the latest tools, technologies, and trends in the industry.
10. **Ethics in Data Science**:
    - Understand and adhere to ethical practices in data handling and analysis.

Remember, there's no single path to becoming a data scientist. The field values skills and practical experience highly, so even non-traditional educational backgrounds can lead to successful careers in data science, provided you have the requisite skills and experience.

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- This is known as **zero-shot prompting**
- Asking the model directly, without examples or structure

4. **Specialize**:
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on platforms like GitHub. This portfolio is crucial

ment:

tend webinars, and participate in hackathons

   - Start applying for data science positions. Tailor your resume and cover letter for each application, highlighting relevant skills and experiences.
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Remember, there's no single pa
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lead to successful careers in da
experience.

Usually results in a very generic output

Zero-Shot prompt - Examples

- E-Learning
  - What is NLP?
  - How to learn Generative AI?
- Coding
    - Python basic examples
- Personal Finance
  - How to earn money in the stock market?
- Health
  - What is a good diet plan?
- Business
  - Write a deadline extension mail to my client.
- Literature
  - Write a poem for my teacher

Zero-Shot prompt – Issues

- E-Learning
  - Explain to me - NLP? - ChatGPT doesn't know whether you are a beginner or an expert
  - How to learn Generative AI?
- Coding
  - Python basic examples –is it for a developer and data scientist?
- Personal Finance
  - How to earn money in the stock market? – Very generic request
- Health
  - What is a good diet plan? – Depends on Gender, Location, Age
- Business
  - Write a deadline extension mail to my client.
- Literature
  - Write a poem for my teacher

Structuring the prompt – 5 Elements

1. Roleplaying : Character Insight/ Persona/ Field Expert!
   - Ask Chat GPT to take the role as an expert
2. Examples : Exceptions / Inclusions and Exclusions!
   - Give some examples of what need to be included and excluded
3. Domain and the Context : Background / Setup/ Context!
   - Explain the context in depth, set the background carefully
4. Display of the output: Size, Structure, and Format of the output!
   - Do you want the output in a tabular format or bullet points or code format
5. Iterations: Refinements/ Regenerations / Improvements
   - Iteration for refining the AI's response to align more closely with the specific requirements

Roleplaying : Character Insight/ Persona/ Field Expert!

- Instruct ChatGPT to step into the shoes of an expert.
- For instance, if your query pertains to coding, request that ChatGPT immerse itself in the role of a seasoned software engineer with a decade of experience.
- Similarly, if you seek advice on a diet plan, prompt ChatGPT to embody the expertise of a certified nutritionist.

Roleplaying : Character Insight/ Persona/ Field Expert!

Basic Prompts	After adding Roleplaying part
Explain to me - NLP?	Put yourself in the shoes of an NLP expert with extensive experience and provide me with an explanation of NLP.
How to learn Generative AI?	Imagine you are an expert in Generative AI and related fields with extensive knowledge. Share insights on how one can effectively learn Generative AI.
Python basic examples	Put yourself in the shoes of a seasoned Python developer with extensive knowledge. Share some foundational Python examples that are essential for beginners to grasp the language effectively.

LAB: Roleplaying

Basic Prompts	After adding Roleplaying part
What is a good diet plan?	
Write a deadline extension mail to my client.	
How to earn money in the stock market?	

LAB: Roleplaying

Basic Prompts	After adding Roleplaying part
What is a good diet plan?	Imagine you are a certified nutritionist with extensive expertise. Provide personalized guidance on what constitutes a highly effective and tailored diet plan
Write a deadline extension mail to my client.	Imagine you are a seasoned business communication expert with years of experience. Craft a professional and persuasive email to request a deadline extension from your client
How to earn money in the stock market?	Imagine you are an experienced stock market expert with a proven track record. Provide valuable insights and strategies on how individuals can effectively earn money in the stock market.

Include some Examples

- Demonstrate to the ChatGPT a few examples as inputs and instruct it to use these examples as a guide to respond to questions.
- This process, known as **few-shot prompting**, involves supplying the AI with a limited number of examples or "shots" that illustrate the task you want it to perform.
- Through this approach, the AI learns and adapts based on these examples.

- For example, if we give two examples of the customer feedback and classify them as positive or negative, the third example will be automatically classified by ChatGPT

Include some Examples = few-shot prompting

Input- Overall okay. Biriyani was not as good. The rest of the items were good. Kadai paneer we found it to be too high a price for the quantity served. Ambiance is nice

Output - Positive - High Price

Two examples(shots) are given to chatGPT

Input- There's no menu card. staff to say the menu card so that the customer feels uncomfortable asking for everything. Overall taste wise just average and overhyped and on overpriced.

Output - Negative - Menu Card issue

Input- I never leave this place disappointed. All the dishes are good and well served. The staff is attentive and suggests good menu options

What is the Output ?

Include some Examples



You

Overall okay. Biriyani was not as good. The rest of the items were good. Kadai paneer we found it to be too high a price for the quantity served. Ambience is nice ---> Positive - High Price

There's no menu card. staff to say the menucard so that the customer feels uncomfortable asking for everything. Overall taste wise just average and overhyped and on overpriced. ---> Negative - Menu Card issue

Inever leave this place disappointed. All the dishes are good and well served. Staff is attentive and suggest good menu options --->





ChatGPT

Positive - Excellent Service and Food Quality

Few Shots - Example

1. Burger
   Harmful Substances: Trans fats, high sodium, refined carbs, artificial flavors.
   Average Calories: 300 - 600 calories (depending on size and ingredients)
   Steps to Burn: Steps required: 7,500 - 15,000 steps.
2. French Fries
   Harmful Substances: Trans fats, acrylamide, high sodium.
   Average Calories: 300 - 400 calories (medium serving)
   Steps to Burn: Steps required: 7,500 - 10,000 steps.
3. Pizza
4. Fried Chicken
5. Sugary Soft Drink
6. Potato Chips
7. Ice Cream
8. Deep Oil fried Samosa

Few Shots - Example

Sure! Here's the extended list continuing from your given format:

3. Pizza

- **Harmful Substances**: Saturated fats, refined carbs, high sodium.
- **Average Calories**: 270 - 350 calories (depending on toppings and crust)
- **Steps to Burn**:
  - **Steps required**: 6,750 - 8,750 steps.

4. Fried Chicken

- **Harmful Substances**: Trans fats, saturated fats, high sodium.
- **Average Calories**: 300 - 400 calories (depending on size and coating)
- **Steps to Burn**:
  - **Steps required**: 7,500 - 10,000 steps.

5. Sugary Soft Drink

- **Harmful Substances**: High fructose corn syrup, artificial colorants, caffeine.
- **Average Calories**: 140 - 150 calories (330 ml can)
- **Steps to Burn**:
  - **Steps required**: 3,500 - 3,750 steps.

6. Potato Chips

- **Harmful Substances**: Trans fats, acrylamide, high sodium, artificial flavorings.
- **Average Calories**: 150 - 170 calories (small bag, 30g)
- **Steps to Burn**:
  - **Steps required**: 3,750 - 4,250 steps.

7. Ice Cream

- **Harmful Substances**: Saturated fats, added sugars, artificial flavorings.
- **Average Calories**: 250 - 300 calories (1 cup)
- **Steps to Burn**:
  - **Steps required**: 6,250 - 7,500 steps.

8. Deep Oil Fried Samosa

- **Harmful Substances**: Trans fats, refined carbs, high sodium.
- **Average Calories**: 200 - 250 calories (per samosa)
- **Steps to Burn**:
  - **Steps required**: 5,000 - 6,250 steps.

LAB - Few Shots

```
Complete the below workout plan for 6 days a week

Day-1: Chest
Push-Ups - 2 Sets - 15 reps
Chest Flyes - 2 Sets - 15 reps
Bench Press - 2 sets - 15 reps

Day-2: Shoulders
```

LAB - Few Shots

```
Complete the below workout plan for 6 days a week
Day-1: Chest
Push-Ups - 2 Sets - 15 reps
Chest Flyes - 2 Sets - 15 reps
Bench Press - 2 sets - 15 reps

Day-2: Shoulders
```

ChatGPT

Certainly, here's a complete 6-day workout plan with a focus on different muscle groups for each day:

Day 1: Chest

1. Push-Ups - 3 Sets - 15 reps
2. Chest Flyes - 3 Sets - 12 reps
3. Bench Press - 3 Sets - 10 reps

Day 2: Shoulders

1. Seated Dumbbell Shoulder Press - 3 Sets - 12 reps
2. Lateral Raises - 3 Sets - 15 reps
3. Front Raises - 3 Sets - 12 reps

Day 3: Back

1. Pull-Ups or Lat Pulldowns - 3 Sets - 12 reps
2. Bent Over Rows - 3 Sets - 12 reps
3. Deadlifts - 3 Sets - 10 reps

Day 4: Legs

1. Squats - 3 Sets - 12 reps
2. Lunges - 3 Sets - 12 reps (each leg)
3. Leg Press - 3 Sets - 12 reps

Day 5: Arms

1. Bicep Curls - 3 Sets - 12 reps
2. Tricep Dips - 3 Sets - 12 reps
3. Hammer Curls - 3 Sets - 12 reps
4. Skull Crushers - 3 Sets - 12 reps

Day 6: Core

1. Planks - 3 Sets - Hold for 30-45 seconds
2. Russian Twists - 3 Sets - 15 reps (each side)
3. Leg Raises - 3 Sets - 12 reps

Day 7: Rest

Domain and Context

Put yourself in the shoes of a seasoned Python developer with extensive knowledge. Give me the learning path for becoming a **Python developer** in five short steps

ChatGPT

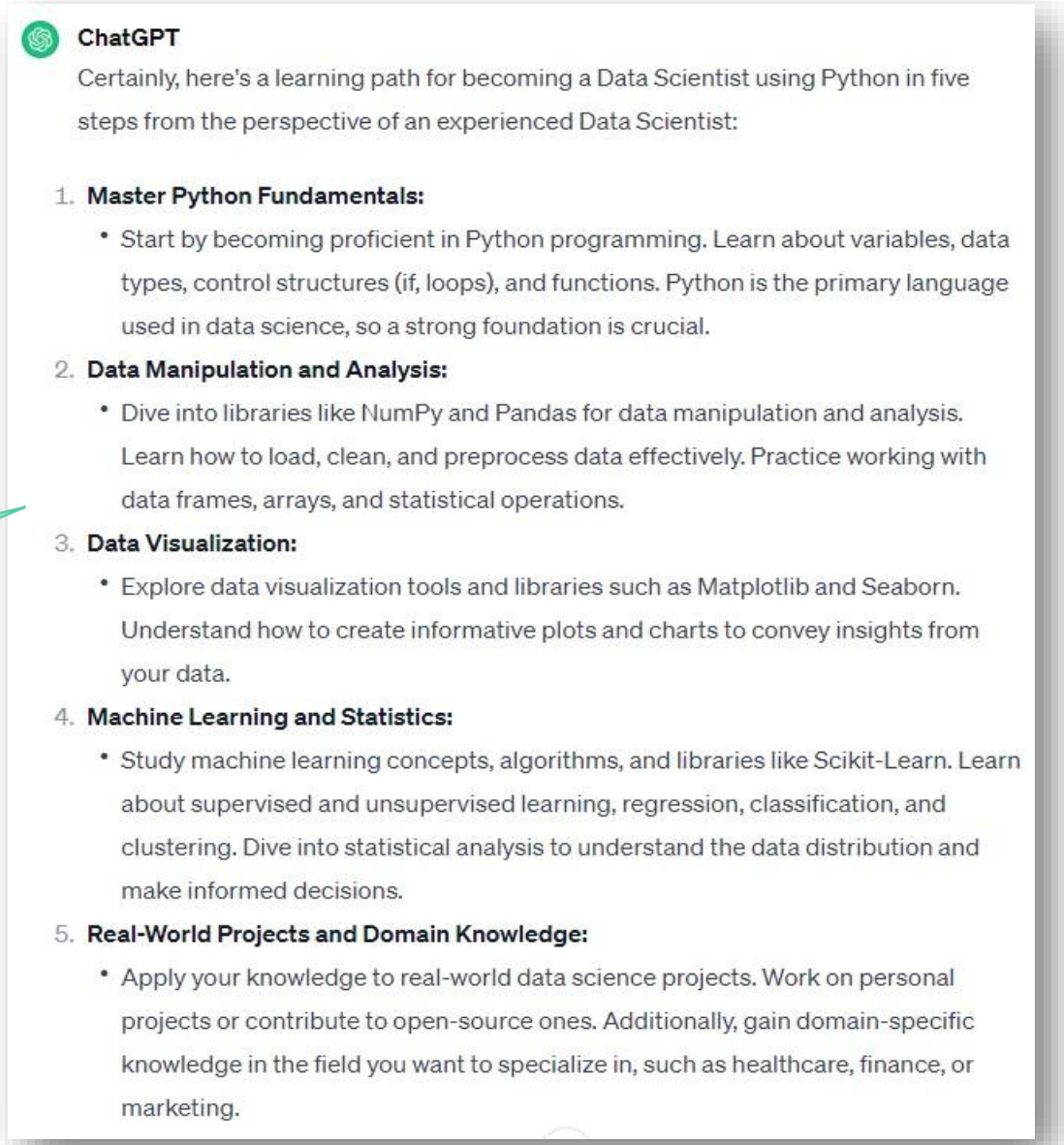
Certainly, here's a learning path for becoming a Python developer in five concise steps from the perspective of an experienced Python developer:

1. **Master the Basics:** Begin with a deep understanding of Python's fundamentals, including variables, data types, control structures (if statements, loops), and functions. Practice coding to build a solid foundation.
2. **Object-Oriented Programming (OOP):** Delve into OOP principles as Python is an object-oriented language. Learn about classes, objects, inheritance, and polymorphism to structure your code effectively.
3. **Data Structures and Algorithms:** Study essential data structures (lists, dictionaries, sets, etc.) and algorithms (sorting, searching) to write efficient and optimized Python code. Practice problem-solving on platforms like LeetCode or HackerRank.
4. **Web Development or Data Science:** Depending on your career goals, choose a specialization path. For web development, explore popular frameworks like Django or Flask. For data science, focus on libraries like NumPy, Pandas, and Matplotlib.
5. **Real-World Projects and Collaboration:** Apply your skills to real-world projects. Contribute to open-source projects, collaborate with other developers, and continuously learn by working on challenging tasks. Stay updated with Python's evolving ecosystem.

Domain and Context

Put yourself in the shoes of a seasoned Data Scientist with extensive knowledge. Give me the learning path for **Python to become a data scientist** in five steps

Compare this output with the previous output



ChatGPT

Certainly, here's a learning path for becoming a Data Scientist using Python in five steps from the perspective of an experienced Data Scientist:

1. **Master Python Fundamentals:**
   - Start by becoming proficient in Python programming. Learn about variables, data types, control structures (if, loops), and functions. Python is the primary language used in data science, so a strong foundation is crucial.
2. **Data Manipulation and Analysis:**
   - Dive into libraries like NumPy and Pandas for data manipulation and analysis. Learn how to load, clean, and preprocess data effectively. Practice working with data frames, arrays, and statistical operations.
3. **Data Visualization:**
   - Explore data visualization tools and libraries such as Matplotlib and Seaborn. Understand how to create informative plots and charts to convey insights from your data.
4. **Machine Learning and Statistics:**
   - Study machine learning concepts, algorithms, and libraries like Scikit-Learn. Learn about supervised and unsupervised learning, regression, classification, and clustering. Dive into statistical analysis to understand the data distribution and make informed decisions.
5. **Real-World Projects and Domain Knowledge:**
   - Apply your knowledge to real-world data science projects. Work on personal projects or contribute to open-source ones. Additionally, gain domain-specific knowledge in the field you want to specialize in, such as healthcare, finance, or marketing.

Domain and Context

Imagine you are a certified nutritionist with extensive expertise. Provide personalized guidance on what constitutes a highly effective and tailored diet plan. **I live in South India. I am a Male. My age is 38. My height is 5.8 and My weight is 73kg.**

Compare this output with the previous output

As a certified nutritionist, I can provide you with some personalized guidance for a highly effective and tailored diet plan based on your age, gender, location in South India, height, and weight. However, please note that it's always best to consult with a healthcare professional or registered dietitian for a comprehensive and individualized plan. Here's a general diet plan to get you started:

Breakfast:

- Start your day with a nutritious breakfast.
- Include options like idli, dosa (preferably whole-grain), poha, or upma made with minimal oil.
- Add a side of sambar or coconut chutney for flavor and added nutrients.
- Incorporate a serving of fresh fruits, like a banana or papaya.

Mid-Morning Snack:

- Have a small portion of nuts or seeds for a healthy snack. Almonds or walnuts are good choices.
- You can also enjoy a cup of buttermilk or coconut water.

Lunch:

- Opt for a balanced lunch that includes whole grains, lean protein, and plenty of vegetables.
- Have a serving of brown rice or millets like ragi or bajra.
- Include a source of protein like lentils (dal), fish, or lean chicken.
- Load up on vegetables, such as spinach, carrots, and cucumber.
- Use minimal oil and spices in your cooking.
- Finish your meal with a serving of yogurt or curd.

Afternoon Snack:

- A light snack can help keep your energy levels steady.
- Consider having a fruit salad, a handful of roasted chickpeas, or a vegetable soup.

Dinner:

- Keep your dinner relatively light.
- Include a variety of vegetables and a small portion of whole grains.
- Opt for options like vegetable curry with chapati or a simple khichdi.
- Avoid heavy or rich dishes before bedtime.

Before Bed:

- If you feel hungry before bedtime, have a glass of warm milk or a herbal tea like chamomile.

Domain and Context

```
Assume you're an experienced investment banker and stock analyst teaching a class on long-term investing. Could you explain the key factors or ratios that are crucial for evaluating stocks for long-term investment? List these in a short and an easy-to-understand format suitable for beginners.
```

Certainly! Here are the key factors and ratios for evaluating stocks for long-term investment in a concise and beginner-friendly format:

1. **Earnings Per Share (EPS):** Look for growing and consistent profits per share.
2. **Price-to-Earnings Ratio (P/E Ratio):** Lower P/E ratios can indicate better value.
3. **Dividend Yield:** Higher yields provide regular income for investors.
4. **Debt-to-Equity Ratio:** Lower debt levels suggest financial stability.
5. **Return on Equity (ROE):** Higher ROE reflects effective use of shareholder equity.
6. **Market Capitalization:** Consider companies of various sizes based on your risk tolerance.
7. **Competitive Advantage (Moat):** Seek companies with a sustainable edge in their industry.
8. **Management Quality:** Research the leadership team's track record.
9. **Industry and Market Trends:** Understand the industry's growth potential.
10. **Diversification:** Spread risk by investing in different stocks and sectors.
11. **Long-Term Prospects:** Consider a company's ability to adapt and grow.
12. **Valuation Relative to Peers:** Compare with similar companies in the industry.

Domain and Context

Explain the philosophy of BhagwatGeeta to a **12 years old kid in 5 points**

Certainly! Here's an explanation of the philosophy of the Bhagavad Gita in 5 simple points for a 12-year-old

1. **Duty and Responsibility:** The Bhagavad Gita teaches us about the importance of doing our duties and responsibilities sincerely, without worrying too much about the results. It's like doing your homework or chores because it's the right thing to do, not just for rewards or praise.
2. **Inner Peace:** The Gita tells us that finding inner peace and calmness is essential. It teaches us to control our mind and emotions, like handling your feelings when things don't go your way, and finding balance in life.
3. **Self-Discovery:** It encourages us to understand who we are and what our true self is. This means knowing your strengths, weaknesses, and values, which helps make better decisions in life.
4. **Equality and Respect:** The Gita teaches that all people are equal and should be treated with kindness and respect, regardless of their background or appearance. It's like being friendly to everyone at school.
5. **Spirituality:** Finally, the Gita talks about connecting with a higher power or inner spirituality. It's like feeling a sense of wonder about the world and understanding that there's something bigger than ourselves.

So, the Bhagavad Gita teaches us to do our best, stay calm, understand ourselves, be kind to others, and explore our inner world and spirituality. It's like a guide to living a good and meaningful life.

Display the output

- You Can display the output in many different formats
  - Bullet Points & Numbered Lists
  - Tabular Values
  - Sections and Subsections
  - Code Blocks & Math Equations
  - Quotes or Dialogues
  - Labels and Annotations

Display the output - Lists and Bullet points

```
List the top attractions
in Paris, France,. Give
the list only. Do not
include any explanations.
```



You

List the top attractions in Paris, France,. Give the list only. Do not include any explanations.



ChatGPT

1. Eiffel Tower
2. Louvre Museum
3. Notre-Dame Cathedral
4. Montmartre and Sacré-Cœur Basilica
5. Champs-Élysées and Arc de Triomphe
6. Seine River Cruises
7. Palace of Versailles
8. Musée d'Orsay
9. Luxembourg Gardens
10. The Pompidou Center (Centre Pompidou)

Display the output - Lists and Bullet points

```
List key personality traits
that contribute to success,
and present them in a concise
bullet point format.
```

ChatGPT

Certainly! Here are key personality traits that contribute to success presented in a concise bullet-point format:

- **Determination**: The ability to stay focused on goals and persist through challenges.
- **Resilience**: Bouncing back from setbacks and adapting to adversity.
- **Self-discipline**: The capacity to control impulses and maintain a strong work ethic.
- **Optimism**: A positive outlook on life and the belief in one's ability to achieve goals.
- **Adaptability**: Being open to change and able to adjust to new circumstances.
- **Leadership**: The ability to inspire and influence others to achieve common objectives.
- **Problem-solving**: Analyzing situations and finding effective solutions.
- **Creativity**: Thinking outside the box and generating innovative ideas.
- **Effective communication**: The skill to convey thoughts and ideas clearly and persuasively.
- **Empathy**: Understanding and relating to the feelings and perspectives of others.
- **Time management**: Efficiently utilizing time and prioritizing tasks.
- **Persistence**: Refusing to give up in the face of obstacles.

Display the output - Lists and Bullet points

```
List 5 effective study
techniques for improving
memory retention
```

This is what I do..

ChatGPT

Certainly! Here are five effective study techniques to help improve memory retention:

1. **Active Recall**: This involves actively testing yourself on the material you're studying rather than simply rereading or passively reviewing notes. Try to recall key concepts and facts from memory, which strengthens your ability to remember them later.
2. **Spaced Repetition**: Instead of cramming all your studying into one session, space out your study sessions over time. Review the material at increasing intervals, focusing more on the areas where you need the most reinforcement. This technique leverages the spacing effect, which enhances long-term retention.
3. **Mnemonic Devices**: Mnemonics are memory aids that help you remember information through associations or creative techniques. Examples include acronyms, rhymes, visual imagery, and memory palace techniques. Mnemonics make information more memorable and easier to recall.
4. **Summarization**: After reading or studying a section of material, summarize it in your own words. This forces you to understand the content deeply and condense it into key points. Summarization helps you organize information and reinforce your memory.
5. **Teaching or Explaining to Others**: Teaching or explaining what you've learned to someone else is an effective way to solidify your understanding and memory of the material. When you can articulate complex concepts to others, it demonstrates your mastery of the subject matter.

Display the output

Lists and Bullet points

- “List key personality traits that contribute to success, and present them in a concise bullet point format.”
- "List 5 effective study techniques for improving memory retention."
- "Highlight the key advantages of 5G technology in a bullet-point list."
- "Summarize the main causes of air pollution in a bullet-point list.“
- "Provide a bullet-point list of effective strategies for improving social media engagement.“
- “List the key safety features to consider when purchasing a family car.”

Display the output - Sections and Subsections

```
Create a structured essay with sections and subsections about Generative AI. Include applications and examples to explain a layman with very limited knowledge.
```

Section 1: Introduction

- **1.1 Overview of Generative AI**
  - Generative AI is a subset of artificial intelligence that focuses on machines creating new content, such as images, text, or even music, rather than just analyzing existing data. It has gained immense popularity due to its creative capabilities.

Section 2: How Generative AI Works

- **2.1 Understanding Neural Networks**
  - Generative AI relies on neural networks, which are computer systems inspired by the human brain. These networks consist of interconnected nodes that process information.
- **2.2 Training the Model**
  - Generative AI models are trained using vast datasets to learn patterns and relationships within the data. For instance, to generate human-like text, a model is exposed to large amounts of text data to grasp language structure.
- **2.3 Generating New Content**
  - Once trained, generative AI models can produce novel content by extrapolating from the patterns they've learned. This includes generating realistic images, writing coherent text, or composing music.

Section 3: Applications of Generative AI

- **3.1 Text Generation**
  - Generative AI is used in natural language processing to create human-like text. For example, it can generate news articles, chatbot responses, or even creative writing pieces.

  Example: Chatbots like GPT-3 can provide conversational responses that sound convincingly human.
- **3.2 Image Synthesis**
  - Generative models can generate images that appear to be photographs of real objects or scenes, even if those objects or scenes don't exist in reality.

  Example: DeepDream, a generative AI developed by Google, creates surreal and artistic images by enhancing patterns in existing pictures.
- **3.3 Creative Art and Design**
  - Generative AI has found applications in art and design, assisting artists in creating unique and visually stunning works.

  Example: AI-generated artwork by artists like Mario Klingemann demonstrates the fusion of human creativity with AI capabilities.
- **3.4 Content Generation in Gaming**
  - In the gaming industry, generative AI is used to create vast, immersive worlds and generate in-game content like characters, landscapes, and quests.

  Example: No Man's Sky utilizes generative algorithms to create an almost infinite number of unique planets and creatures within its game universe.

Display the output

Tabular Format

Create 10 MCQ questions on Python pandas. Arrange the questions in a table format with 6 columns. Question, Option A, Option b, Option C , Option D and the Answer

ChatGPT

Here are 10 multiple-choice questions (MCQs) on Python pandas along with four options (A, B, C, D) and the correct answers:

Question	Option A	Option B	Option C
1. What is the primary data structure in pandas for storing and manipulating data?	A. List	B. Array	C. Series
2. Which function is used to read a CSV file into a pandas	A. `` `read_csv()` ``	B. `` `load_data()` ``	C. `` `read_d ``

Display the output

Tabular Format

What's the study plan for someone who wants to be a data analyst? Note all important fields and their specific points, and organize them in a tabular format.

ChatGPT

To become a data analyst, one needs to focus on a range of topics. Here's a structured learning path in a tabular format:

Main Topic	Subtopics
Basic Mathematics	- Statistics
	- Probability
Programming Skills	- Python/R
	- SQL
Data Processing	- Data Cleaning
	- Data Transformation
Data Analysis	- Exploratory Data Analysis
	- Data Interpretation
Data Visualization	- Using Tools like Tableau, PowerBI, or Matplotlib
Business Intelligence	- Understanding Business Metrics
	- Reporting Techniques
Communication Skills	- Presenting Findings
	- Storytelling with Data
Tools and Technologies	- Excel
	- Database M[illegible]agement Systems

Iterations

- Get the initial response, have a look at it
- Resubmit to get better results
- Adjust the prompt to get better results
- Add a condition to get better results
- Give some more context to get the better results

Iterations

- Most of the LLMs are chatbots- Use the Question and Answer Chat options iteratively to improve the output
- Each **iteration helps refine the response**, making it more accurate and relevant to the user's query or requirement.
- **Iterative prompting** allows you to tailor the information to the specific user, ensuring it's understandable and relevant to them.
- AI models, no matter how advanced, can sometimes provide incorrect or incomplete information.
- Through iterative prompting, you can correct these errors, ask for clarifications, and ensure the accuracy of the information provided.

Structuring the prompt – 5 Elements

R : **Role-player or Expert** : Persona/Roleplaying / Field Expert

E : **Examples** : Exclusions /Exceptions / Inclusions and Exclusions

D : **Domain and Context :** Background / Setup

D : **Display and Format:** Size, Structure and Format of the output

I : **Iterations:** Refinements/ Regenerations / Improvements

To change the context – Ignore previous chats

1. "Clear previous context and start fresh."
2. "Ignore previous messages and focus on the following instructions."
3. "Begin a new conversation with the following questions:"
4. "Reset the chat and provide responses based on the following input:"
5. "Let's start from scratch with the following discussion topics:"

